Aquaponic Greenhouse

Agenda

Overview

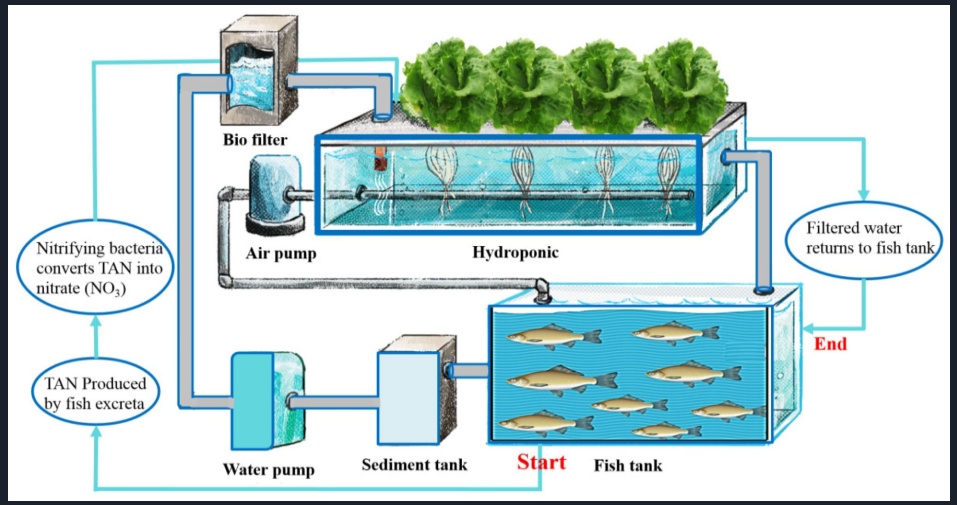
Understanding the problems

Project objective

Demo

Overview

- Self Sustaining ecosystem
  - Fish waste provides nutrients for plants
  - Plants filter the water back to fish



Understanding the problem

01 Lack of Environmental Awareness

- Unpredictable plant growth and tank instability
- Tank and Grow Bed water level monitoring

02 No reliable Feeding Automation

- Flawed Design, only had a version 1.0
- Overfeeding → Dirty water
- Underfeeding → Stressed Fish

03 Detective plant nutrient deficiencies are expensive

- Nitrogen, Phosphorus, Potassium

Addressing the Problems Encountered

01 Lack of Environmental Awareness

- Implement AI/Video Monitoring of Tank Health
- Add Level Switch to the fish tank

02 No reliable Feeding Automation

- Alter Design (Ratcheting Gear), change Motor (Stepper Motor)
- AI/Algorithm type feeding

03 Detective plant nutrient deficiencies are expensive

- Have more money/funding to purchase

Project Features and Future Works

01 Environment Monitoring (Temperature, UV, Light, Water Level)

- Future: Set in more controlled environment where these variables can be controlled (heater, UV lights)

02 Auto Feeder with HMI to send signal to feed fish

- Future: Collect feeding data to determine the ideal amount and timing of food for healthier fish

03 AI monitoring nutrition of plants (Nitrogen, Phosphorus, Potassium)

- Future: Integrate AI predictions into the controlled environment for closed-loop automation.

DEMO